

Your Data, Your Discoveries

PS50008A Week 12 | Research Design Framework

Gordon Wright

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Welcome to Week 12

Who Are We?

Dr Gordon Wright

Module Coordinator

Research: Social cognition, deception, dark personalities and interpersonal perception

Office hours: Thursdays 12:00-13:00

Dr Bence Palfi

Lecturer in the department and our second Lab instructor

Research: AI, Human-Computer Interactions, Meta-Science

What Is This Module?

Research Methods & Experimental Design

How psychologists:

- Ask questions about the mind and behaviour
- Design studies to answer those questions
- Collect and analyse data
- Draw valid conclusions

Why Research Methods?

The Problem with Common Sense

- “Opposites attract”
- “Birds of a feather flock together”
- Both can’t be true... can they?

Common sense often gives contradictory answers. Science provides a systematic way to find out what’s actually true.

Everyday Thinking vs Scientific Thinking

Everyday Thinking

- Based on personal experience
- Influenced by biases
- Accepts anecdotes as evidence
- Confirmation bias
- “I just know”

Scientific Thinking

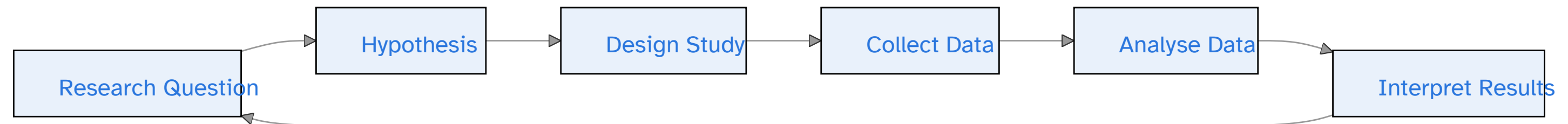
- Based on systematic observation
- Controls for biases
- Requires replicable evidence
- Seeks disconfirmation
- “Let’s test it”

Why Should You Care?

- Evaluate claims you encounter daily
- Understand how psychological knowledge is created
- Essential for your dissertation
- Transferable skills for any career
- Required for postgraduate study

The Research Process

From Question to Knowledge



Step 1: The Research Question

Where do research questions come from?

- Curiosity about behaviour
- Gaps in existing research
- Real-world problems
- Theory testing
- Unexpected observations

Example: Do people lie more when texting than face-to-face?

Step 2: The Hypothesis

A **hypothesis** is a specific, testable prediction about what you expect to find.

Good hypothesis:

“Participants will tell more lies per conversation when communicating via text message compared to face-to-face.”

Not a hypothesis:

“I wonder if texting affects lying.”

Step 3: Design the Study

Key decisions:

- Who will you study? (participants)
- What will you measure? (variables)
- How will you measure it? (methods)
- What comparisons will you make? (design)

We'll spend most of this module on this step.

Step 4: Collect Data

- Follow your plan precisely
- Treat participants ethically
- Record everything carefully
- Minimise errors and bias

Step 5: Analyse Data

Statistics are tools for finding patterns in data and determining whether those patterns are meaningful.

This is where **R** comes in — and what you'll learn in the labs.

Step 6: Interpret Results

- What do the findings mean?
- Do they support the hypothesis?
- What are the limitations?
- What questions remain?

Then the cycle continues...

What You'll Learn

Project Timeline

Week	Milestone
11	See how much fun collecting data is!
12-15	Develop research question and design
—	<i>Reading Week</i> — Work on Research Plan
16	Research Plan due ; Begin data collection
17-18	Data collection and initial analysis
19-20	Complete analysis; Develop write-up (term ends)
Spring Break	Individual report writing
Post-term	Research Report due Friday 1 May

Lectures vs Labs

Lectures (Wednesday 10-12 from week 12)

- Concepts and theory
- Research design principles
- How to think about data

Labs (Thursday 10-12)

- Working with real data
- Building practical skills
- Applying design principles
- Learning about statistical test with R

They work together — lecture concepts, lab application.

Why R? (nothing to fear!)

- Free and open source
- Industry standard in data science
- Powerful data analysis and visualisation
- Reproducible research
- Employable skill

No Experience Required

We assume you've never programmed before. We start from zero.

Assessment Overview

Find an [overview here](#)

Assessment	Weight	Deadline	Week
Research Plan	20%	Fri 27 Feb 2026, 12 noon	16
MCQ Exam	20%	Wed 11 Mar 2026, in lecture slot	18
EDS Assignment	20%	Fri 20 Mar 2026, 12 noon	20
Research Report	40%	Fri 1 May 2026, 12 noon	Post-term

Before We Begin



Bookmark These

- **Glossary** — Look up any unfamiliar terms
- **Self-Assessment Quizzes** — Check your understanding
- **Textbook Chapters** — Go deeper on any topic
- **Hypothes.is** — Ask questions, see classmates' annotations
- **Assessment 1 Guidance** — Start here for your Research Plan

Your data tell a story — let's learn how to read it

Today's Learning Objectives

By the end of this lecture, you will be able to:

- Identify **Independent Variables (IV)** and **Dependent Variables (DV)** in any study
- Distinguish between **between-subjects**, **within-subjects**, and **correlational** designs
- Write **one-tailed** and **two-tailed** hypotheses
- Calculate and interpret **mean** and **standard deviation**

The Big Idea

! Critical Week

Today introduces **ALL** the terminology you need for your Research Plan.

Everything connects: Research Question → Variables → Design → Hypothesis

Act 1: What Are Data?

You generated these data (15 min)

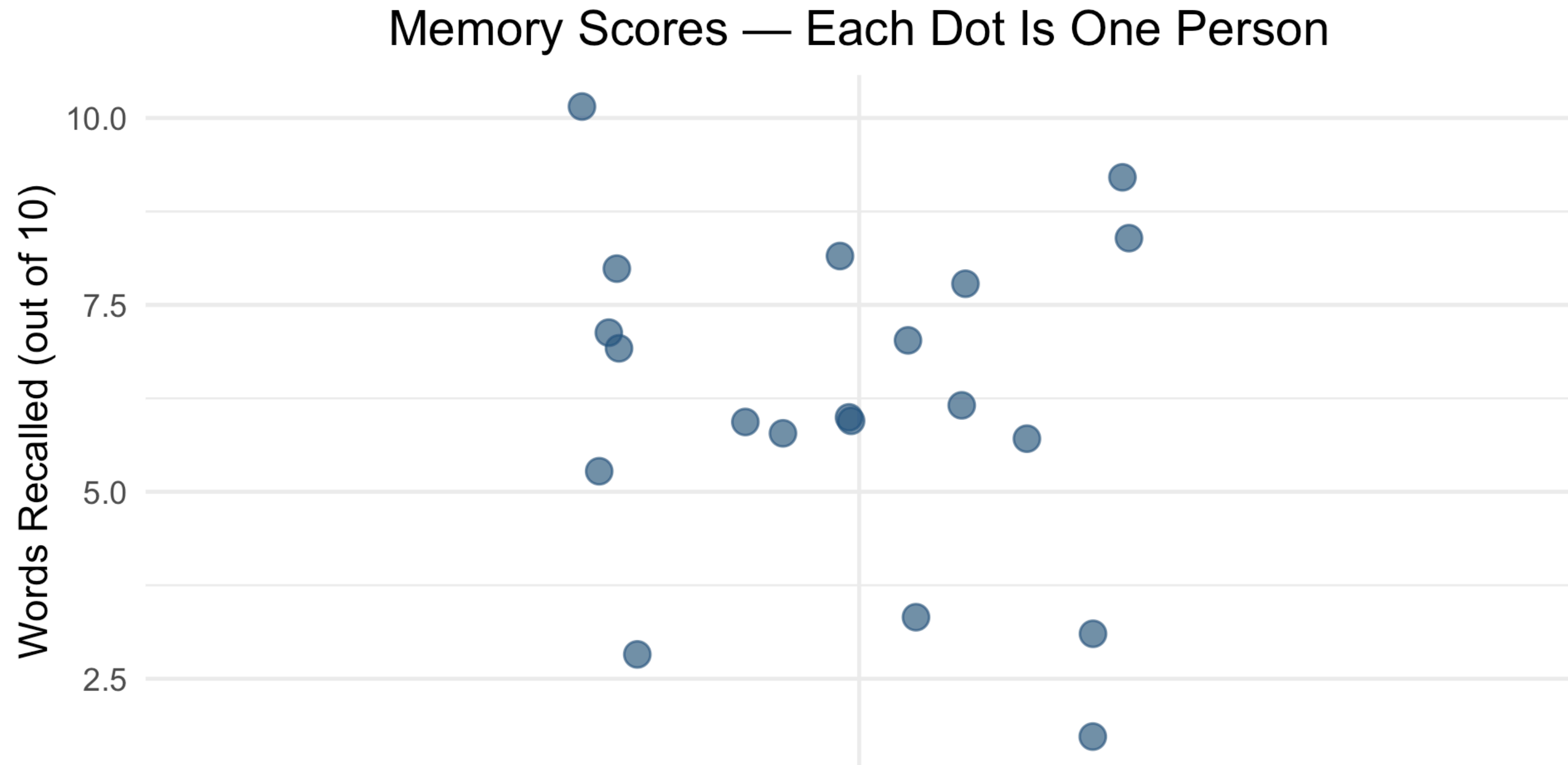
What Did You Actually Do Last Week?

You completed **6 tasks** plus pre/post surveys:

1. **Memory Test** (Mind Palace)
2. **Reflex Test** (Ruler catch)
3. **Time Warp** (60 second estimation)
4. **Sixth Sense** (Staring detection)
5. **Lady Luck** (Random numbers)
6. **Frontal Lob** (Throwing accuracy)

Each task generated NUMBERS — and numbers are what we analyse!

You Are a Data Point - a datum

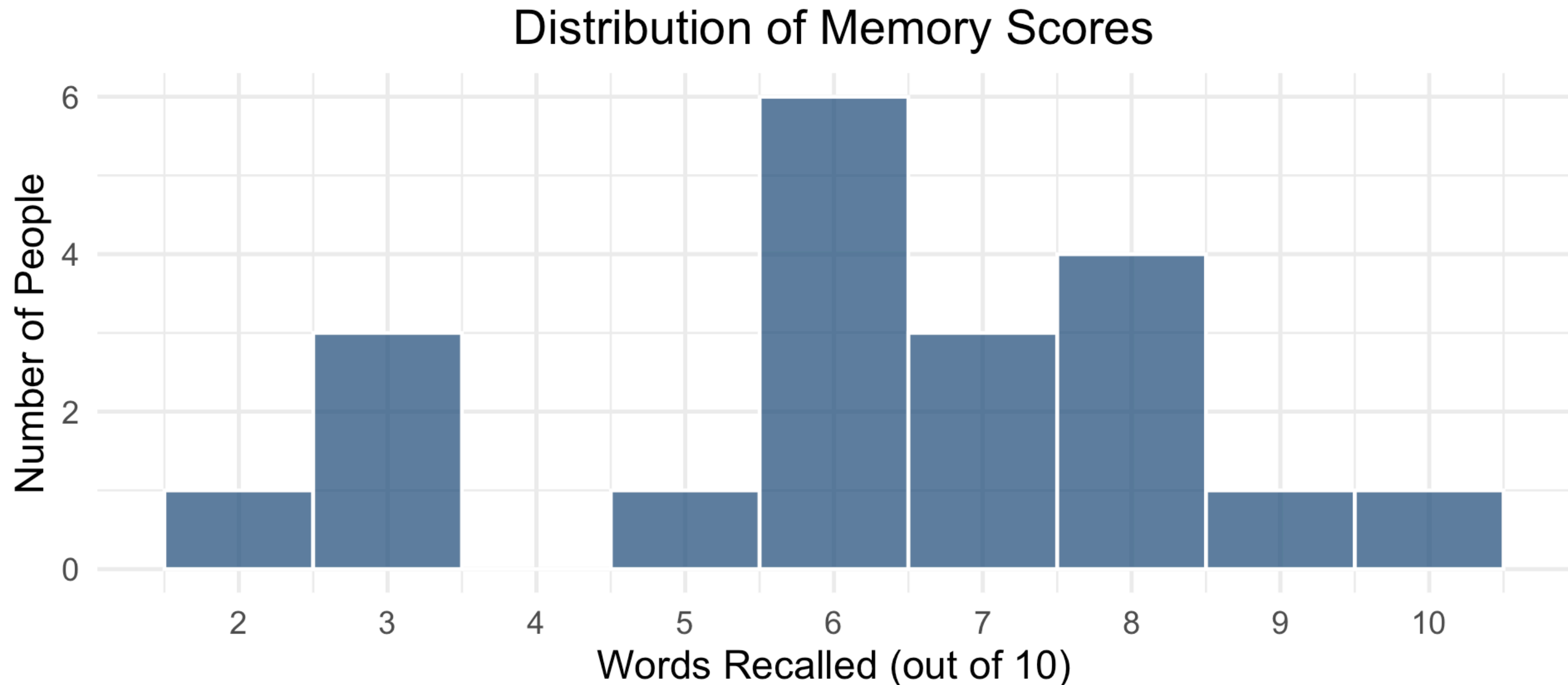


Can you spot yourself?

Act 2: What's Typical?

The simplest question (20 min)

When You Have Numbers, You Want to Know: What's Normal?



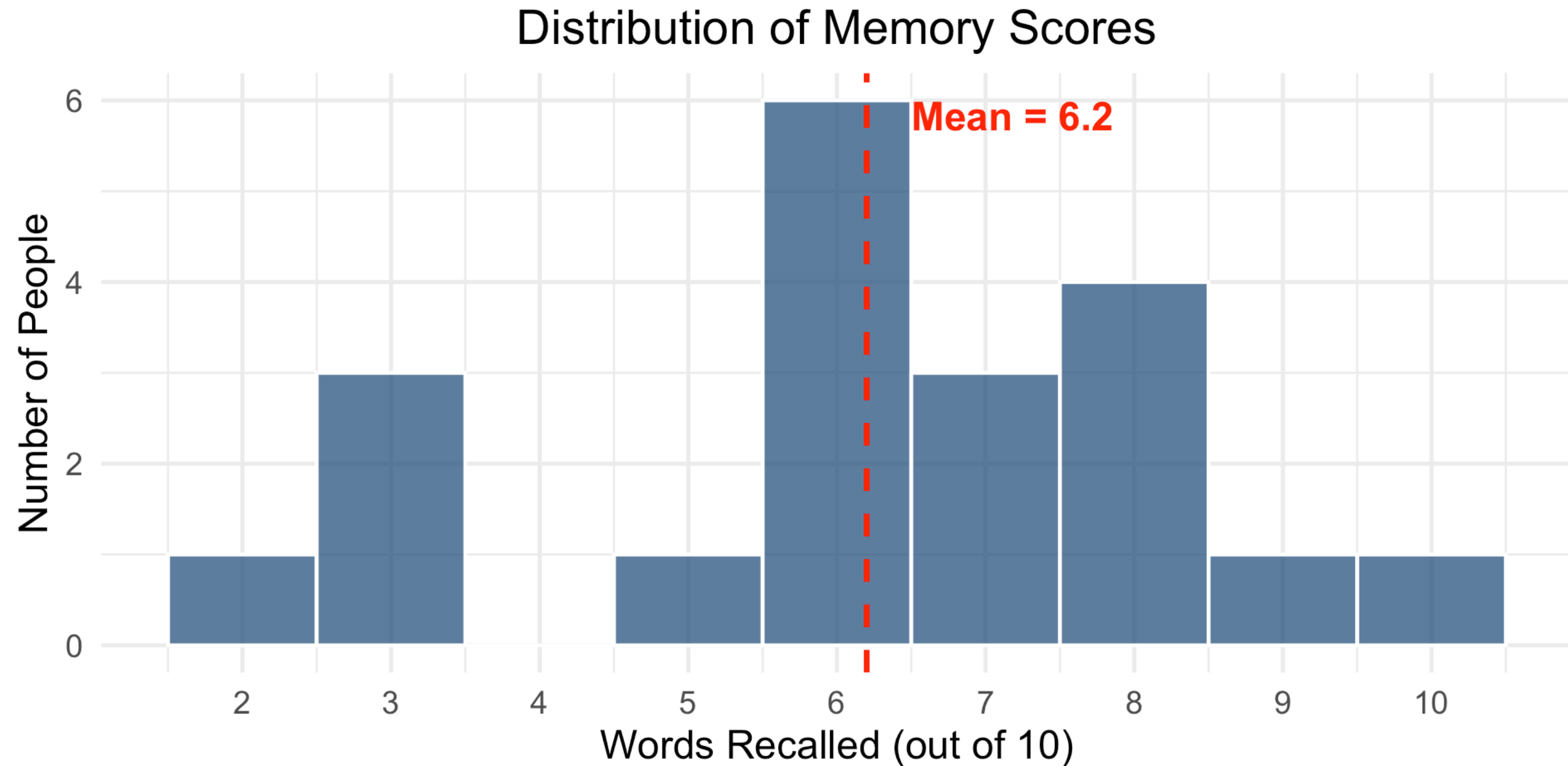
The **histogram** shows how many people got each score.

The Mean: What's Average?

Mean ($\bar{x}$) = Sum of all values $\div$ Number of values

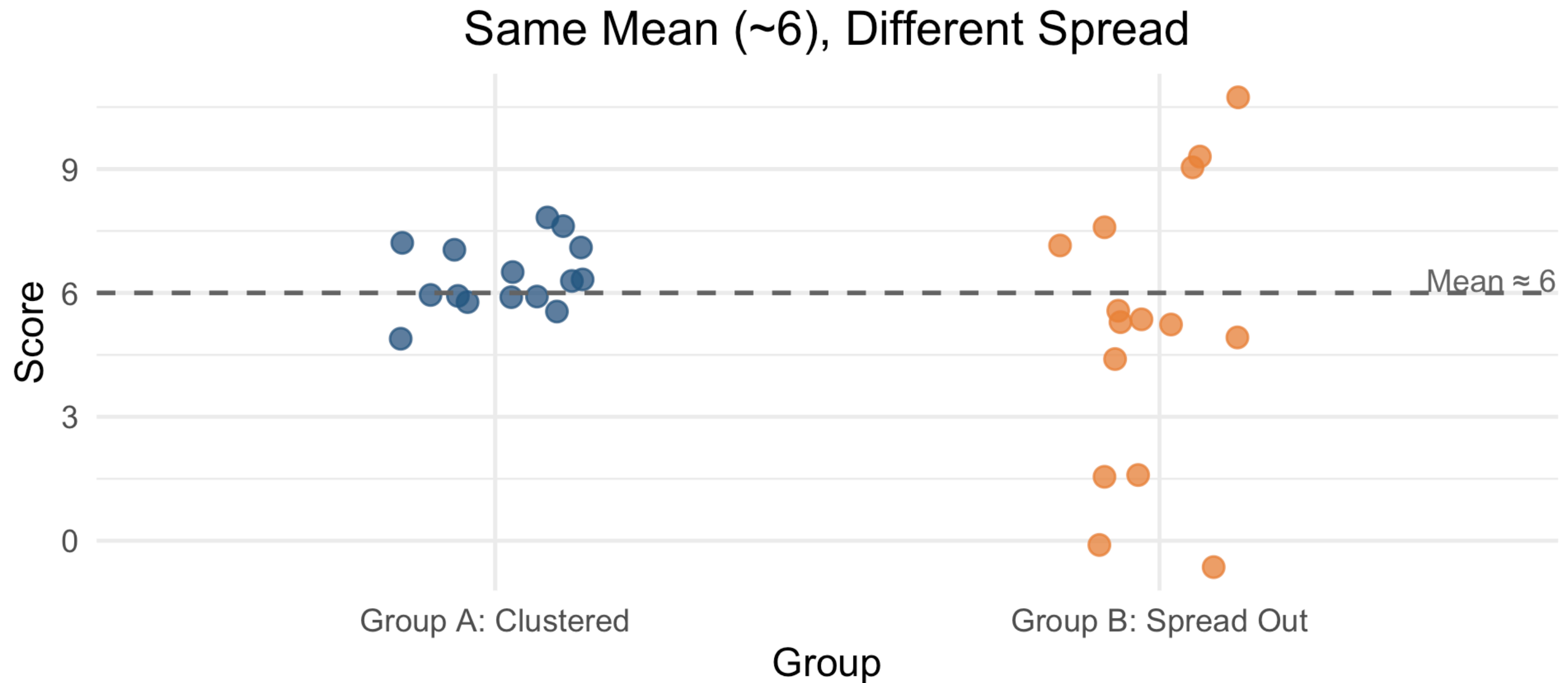
$$\bar{x} = \frac{\sum x_i}{n}$$

That's what I mean!



For your memory scores: **Mean = 6.2 words**

But Wait... Same Mean, Different Stories



Same mean, but Group B is much more variable!

We Need to Measure Spread

Standard Deviation (SD) = How spread out the data are from the mean

- **Low SD** = Scores clustered together (consistent)
- **High SD** = Scores spread out (variable)

For your memory scores: **SD = 2.14 words**

The Key Insight

⚠ Remember This

A **difference between means** only matters if it's **big relative to the spread**.

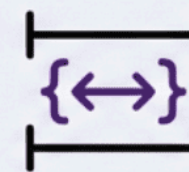
- **Signal** = difference between means
- **Noise** = variability within groups
- The question: Is the signal **bigger** than the noise?

Descriptive Statistics & Formatting



Mean

The arithmetic average.



Standard Deviation

How spread out the data is.

APA Standard Reporting Format:

> $M = x.xx, SD = x.xx$

Annotations:

- Italicize (pointing to M)
- Italicize (pointing to SD)
- Two decimal places (pointing to the $.xx$ in both values)

Precision in formatting conveys professionalism.

Act 3: The Big Reveal

Lab A vs Lab B (25 min)

There Was a Hidden Manipulation

Last week you did the **Memory Test** (Mind Palace).

But here's what you didn't know...

The Secret

Lab A and **Lab B** saw **different words**!

The Manipulation

Lab A (Gordon)

Memorised **abstract** words:

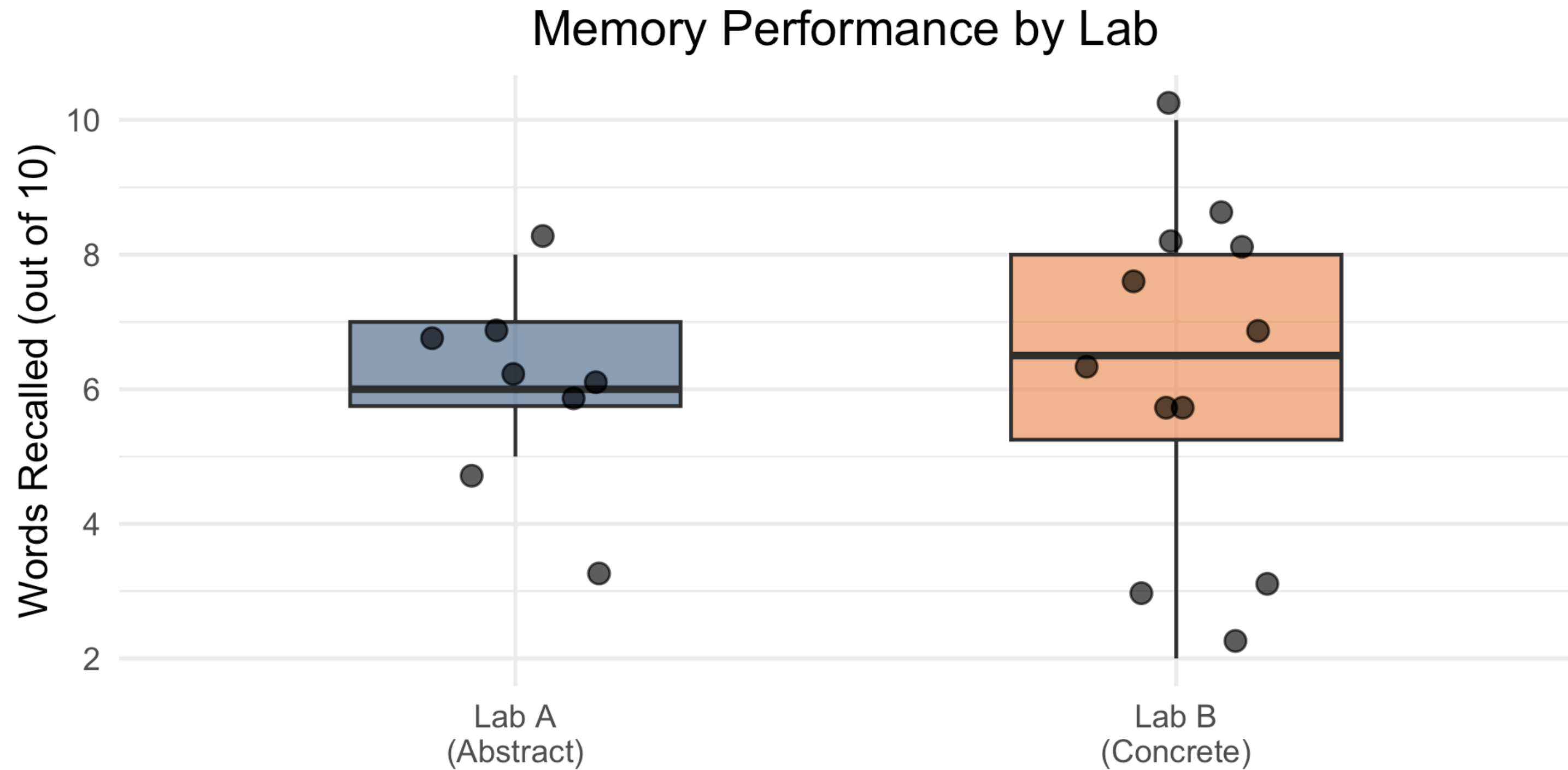
- Justice
- Freedom
- Wisdom
- Courage
- Truth

Lab B (Bence)

Memorised **concrete** words:

- Chair
- River
- Mountain
- Apple
- Hammer

Let's See the Results!



The Numbers

Lab	Word Type	n	Mean	SD
Lab A	Abstract	8	6	1.51
Lab B	Concrete	12	6.3	2.53

Difference: 0.3 words

Is this difference **real** or just **chance**?

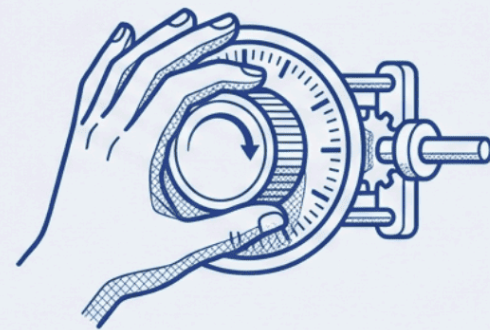
Join us in week 14 and we will put that to the **t-test**!

Now Let's Introduce the Terminology

Independent Variable (IV): What the researcher **manipulates** or controls

Dependent Variable (DV): What the researcher **measures** — the outcome

The Core Elements of Investigation



Independent Variable (IV)

The Driver. What the researcher manipulates or selects.



Dependent Variable (DV)

The Outcome. What the researcher measures.

The fundamental dynamic of an experiment is observing how the manipulation of the IV impacts the measurement of the DV.

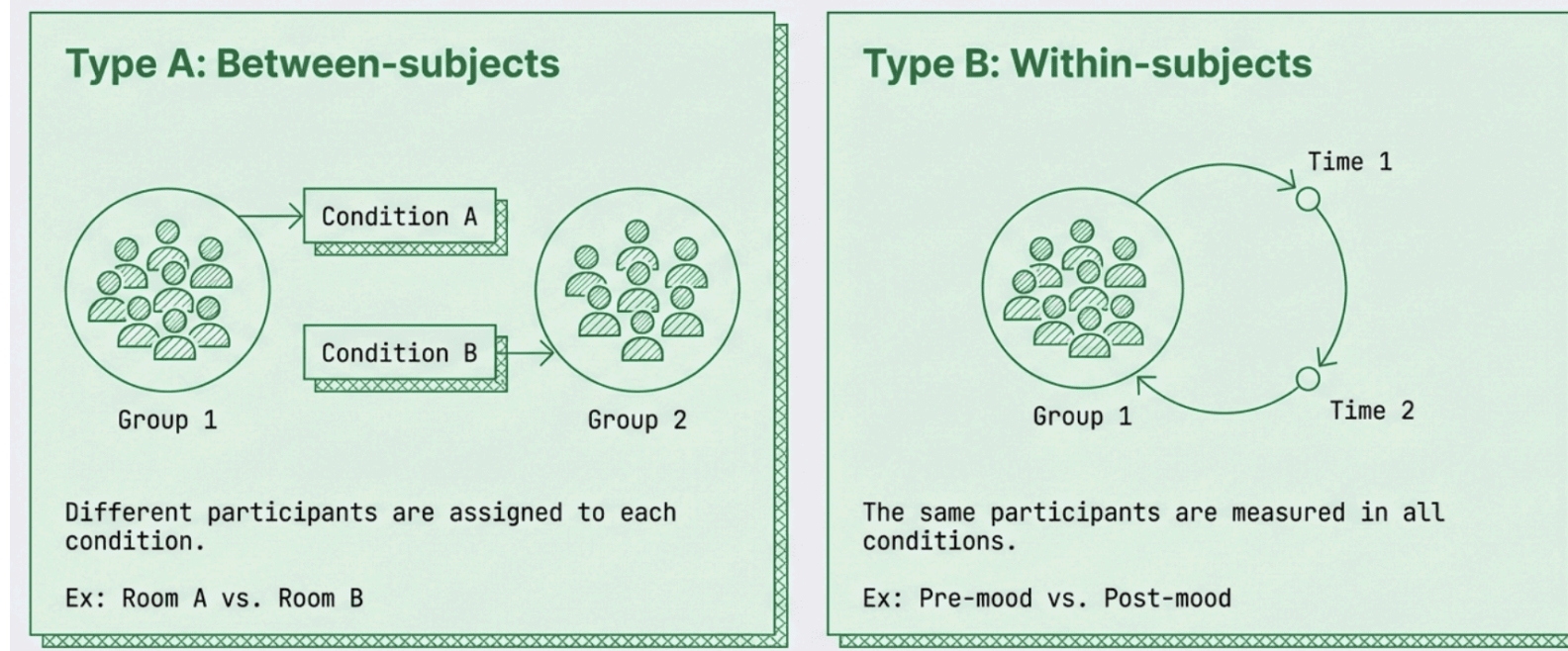
In the memory experiment:

- **IV** = Word type (abstract vs concrete)
- **DV** = Number of words recalled (0-10)

Design Type: Between-Subjects

Between-subjects design: Different participants in each condition

Structural Frameworks: Grouping Participants



- Lab A people **only** saw abstract words
- Lab B people **only** saw concrete words
- You couldn't be in **both** labs!

Why? **Between** = different people in each group

Writing a Hypothesis

Experimental hypothesis (H_1): Predicts an effect or difference

Null hypothesis (H_0): Predicts no effect or no difference

For the memory experiment:

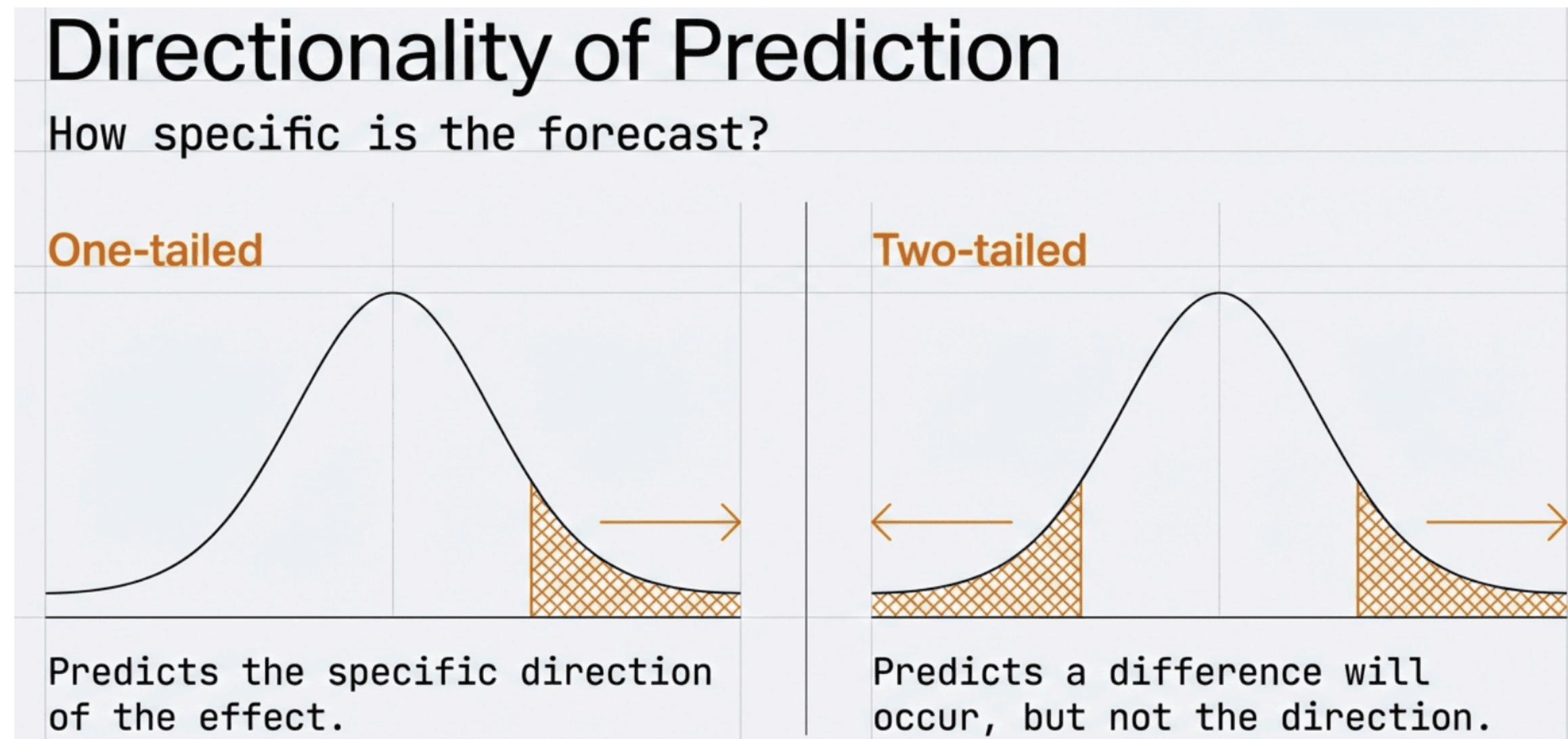
- **H_1 :** Concrete words will be easier to remember than abstract words
- **H_0 :** There will be no difference in recall between word types

One-Tailed vs Two-Tailed

One-tailed: Predicts the **direction** of the effect

Two-tailed: Predicts a difference but **not** which direction

- **One-tailed:** “Concrete words will be **better** than abstract” (predicts direction)
- **Two-tailed:** “Word type will **affect** memory” (no direction)

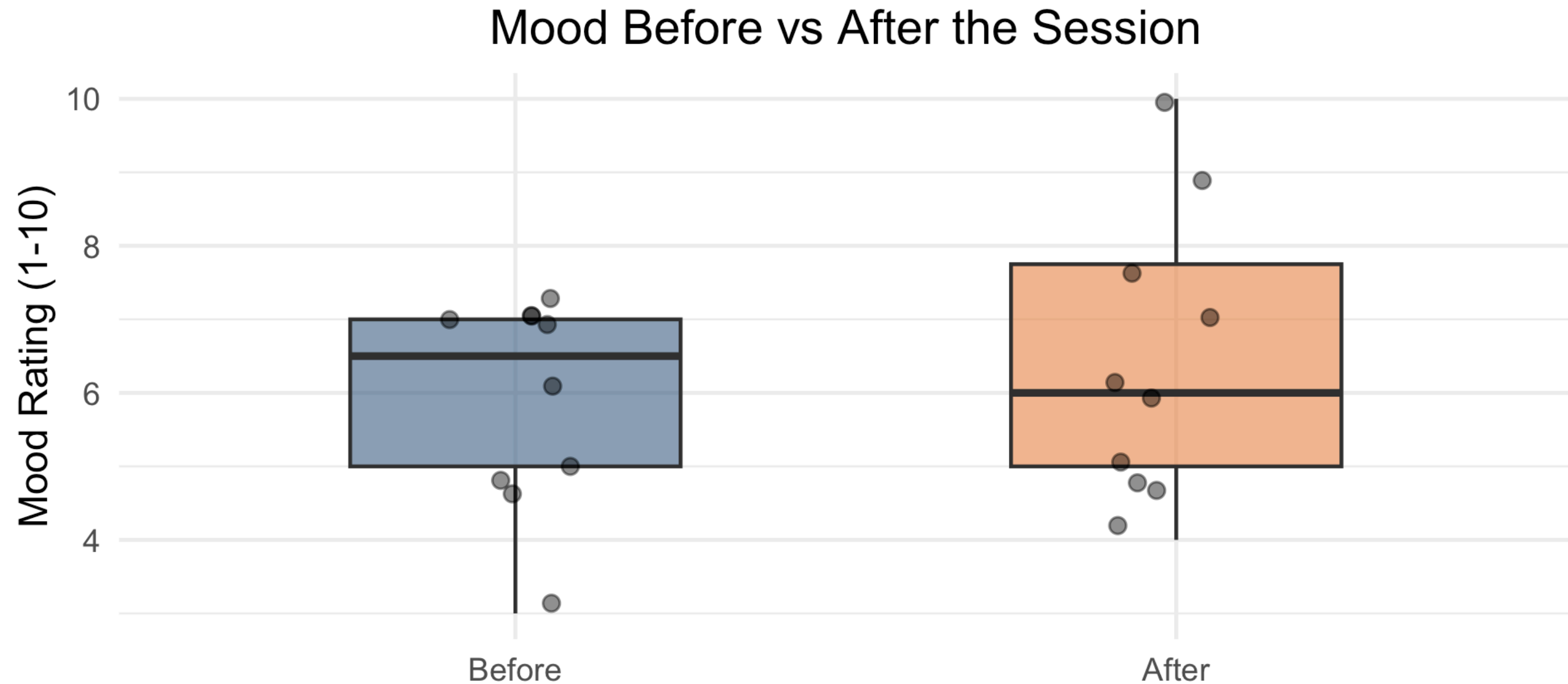


Act 4: Same Person, Different Conditions

Within-subjects design (20 min)

The Pre-Post Mood Survey

Remember you rated your **mood** at the start AND end of the session?



This Is Within-Subjects Design

Within-subjects design: The **same participants** measured in **all conditions**

- Pre-mood and post-mood came from **the same person**
- You're being compared **to yourself**

Why Is This Powerful?

The Power of Within-Subjects

Each person is their own **control**!

Individual differences are **controlled**:

- If your pre-mood was 5 and post-mood was 7...
- **You** improved by 2 points
- We don't need to worry about other people's baseline moods

Act 5: Relationships, Not Differences

Correlational design (15 min)

Self-Rating vs Actual Performance

Before the throwing task, you **rated** how good you thought you'd be.

Question: Do people know themselves?



This Is Correlational Design

Correlational design: Measuring the **relationship** between two variables — **no manipulation**

- **Variable 1:** Self-rating (1-10)
- **Variable 2:** Actual accuracy (%)

 Key Point

There is **no IV** in a correlational design!

The Critical Warning

⚠ Say It With Me

Correlation ≠ Causation

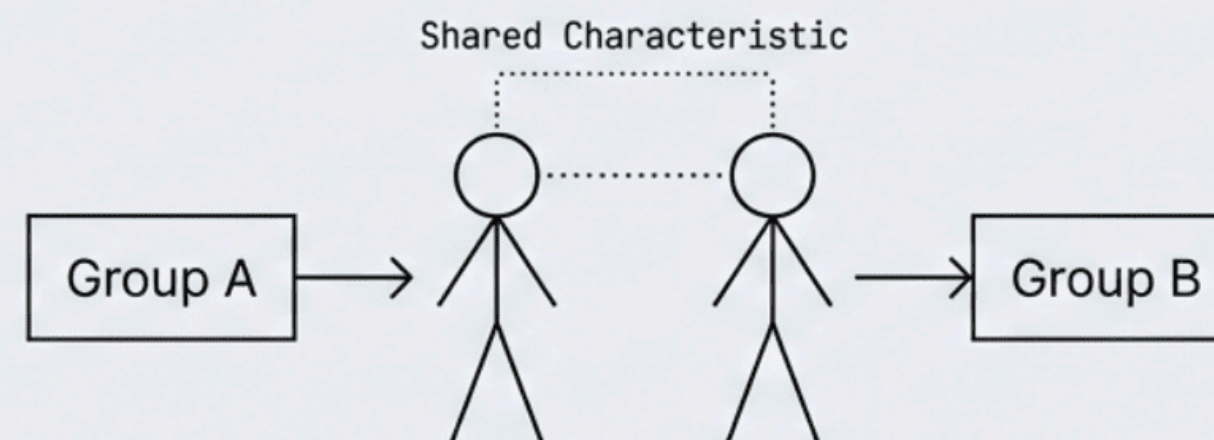
Finding a relationship does NOT mean one variable **causes** the other.

Even if self-ratings predict performance, we can't say self-ratings **cause** good performance.

Specialized Configurations

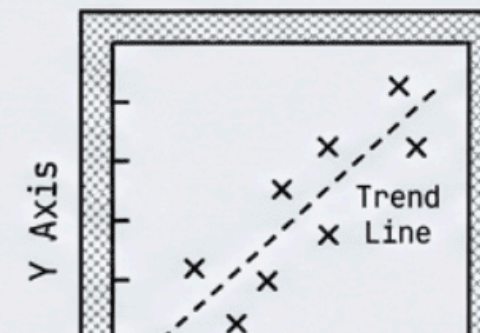
Matched Pairs

Participants are paired based on key characteristics before assignment.



Correlational Design

Measuring relationships between variables (no manipulation).



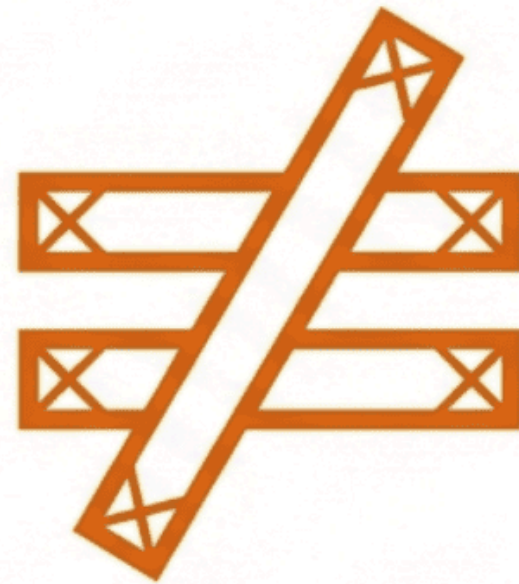
Act 6: Was It Just Luck?

The null hypothesis (20 min)

Predicting Reality

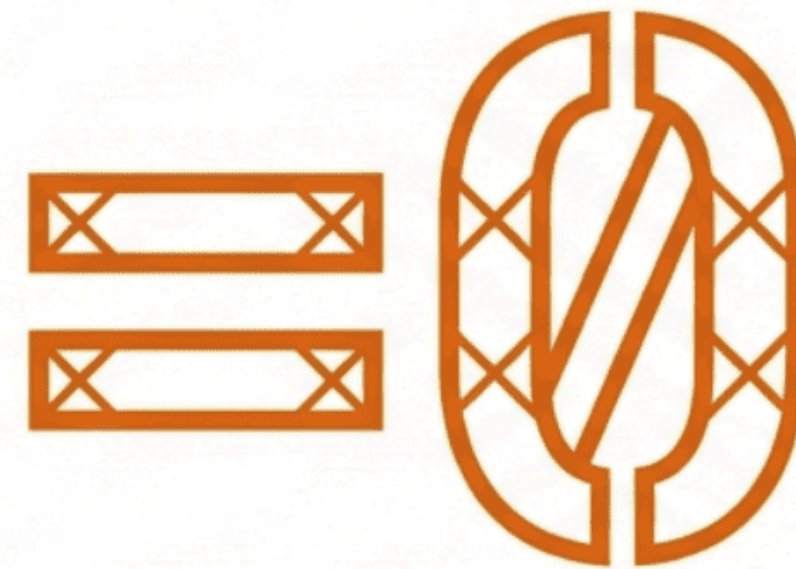
Research is a contest between two possibilities.

The Experimental Hypothesis



Predicts an effect or relationship will exist. Something happens.

The Null Hypothesis



Predicts no effect or relationship. Nothing changes.

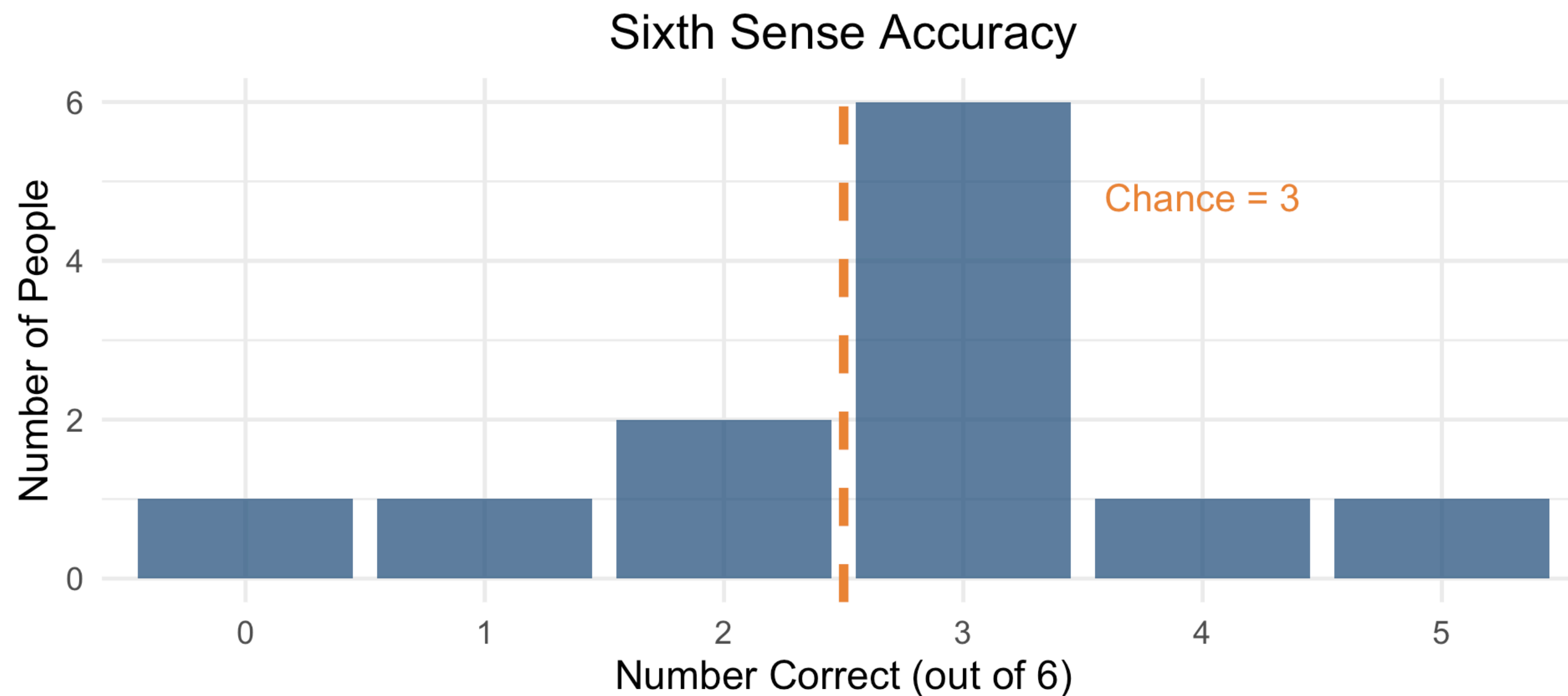
The Sixth Sense Task

You tried to detect if someone was staring at you.

- 6 trials
- Guessed “staring” or “not staring”

If you're just guessing: You'd expect **3 out of 6** right (50%)

Your Results



Some People Got 5/6 or 6/6!

The Question

Does getting 5 or 6 right mean you have a **sixth sense**?

Not necessarily!

If you flip a coin 6 times, sometimes you get 5 or 6 heads **just by luck**.

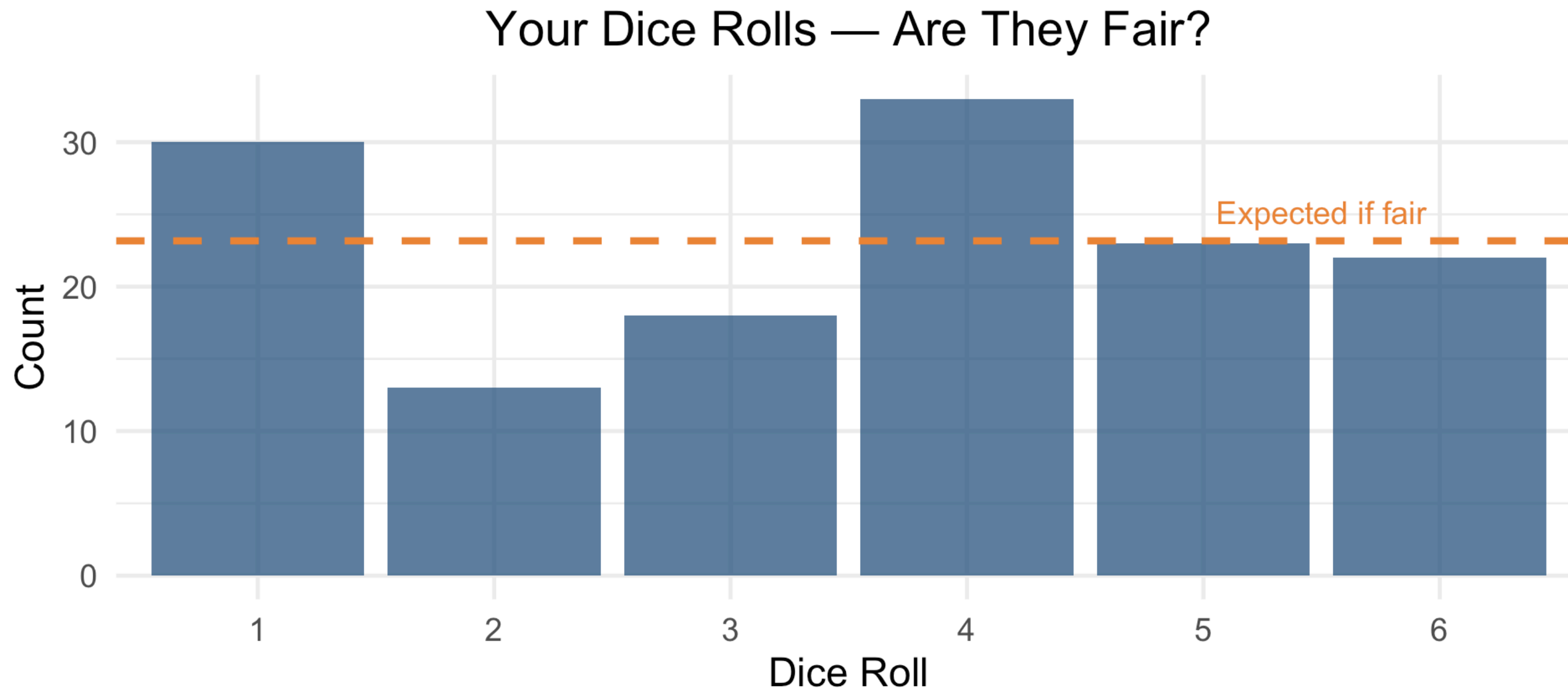
The Null Hypothesis

Null hypothesis (H_0): The result is due to **chance**, not a real effect.

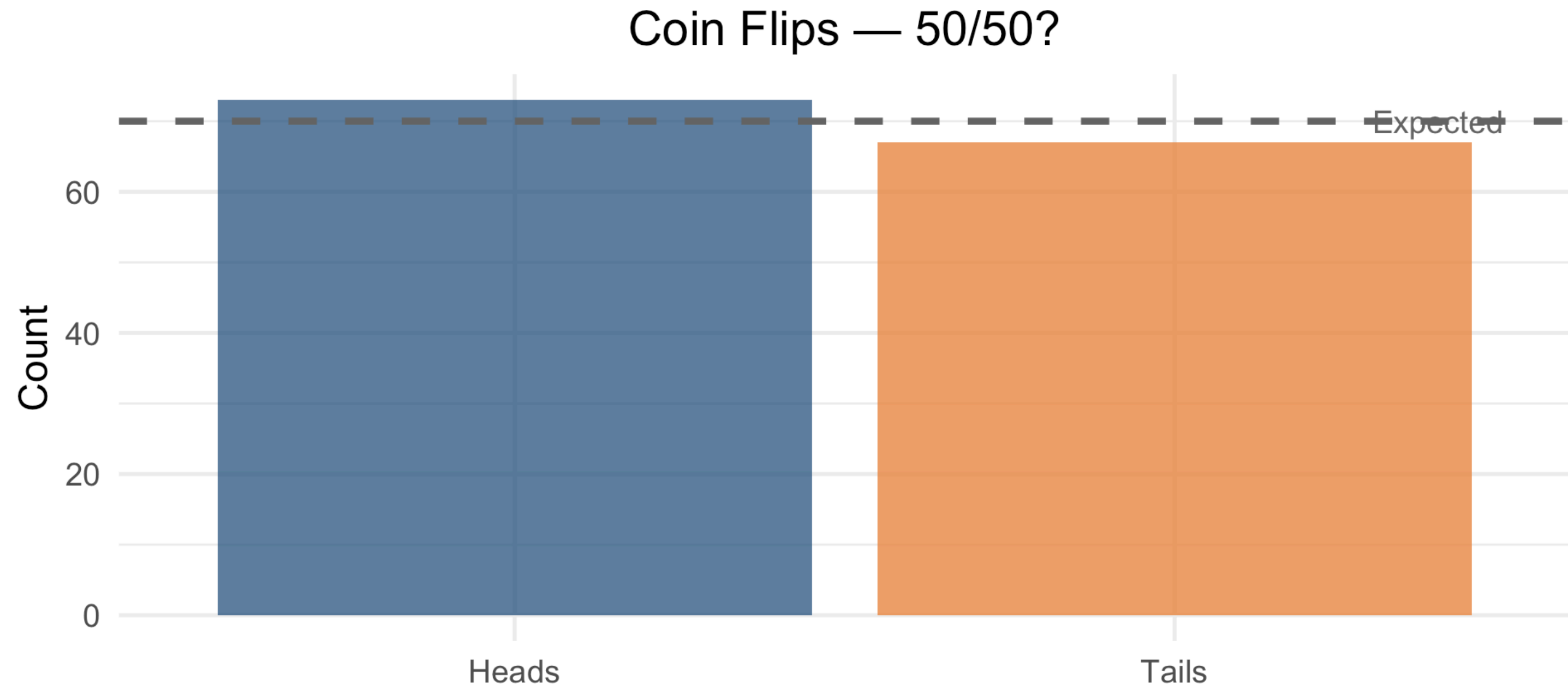
For the sixth sense:

- **H_0 :** People are just guessing (performance = 50%)
- **H_1 :** People can detect staring (performance $\neq$ 50%)

Your Dice Rolls



Your Coin Flips



Why Statistics?

With 20 people, **some will get lucky**:

- A few might get 5/6 or 6/6 just by chance
- Dice won't land perfectly evenly
- Coins won't be exactly 50/50

Statistics helps us ask: How surprising is this result? See week14!

The Framework

Putting it all together (5 min)

The Research Design Decision Tree

Step 1: Is there manipulation?

- **Yes** → Experimental design
- **No** → Correlational design

Step 2: If experimental, same or different people?

- **Same people** → Within-subjects
- **Different people** → Between-subjects

The Quick Check

No IV? → Correlational

IV + same people? → Within-subjects

IV + different people? → Between-subjects

DataLab Summary

Task	Design	IV	DV
Memory	Between	Word type	Recall (0-10)
Pre/Post Mood	Within	Time	Rating (1-10)
Frontal Lob	Within	Hand	Accuracy (%)
Sixth Sense	vs Chance	—	Hits (0-6)
Self vs Actual	Correlational	—	—

The Vocabulary You Now Have



Check These Off

- ☒ Independent Variable (IV)
- ☒ Dependent Variable (DV)
- ☒ Between-subjects design
- ☒ Within-subjects design
- ☒ Correlational design
- ☒ One-tailed hypothesis
- ☒ Two-tailed hypothesis
- ☒ Null hypothesis (H_0)
- ☒ Mean and Standard Deviation

Connecting to Assessments

Assessment	What You Learned Today
Research Plan	ALL terminology — you can start planning!
MCQ Exam	Design identification, IV/DV, hypothesis types
EDS Assignment	Calculating means and SDs, identifying designs
Research Report	Method section structure

What's Next?

Thursday Lab: Apply this framework to YOUR DataLab data

- Identify designs, IVs, DVs
- Calculate means and SDs
- Practice writing hypotheses

Focus on conceptual understanding!

Questions?

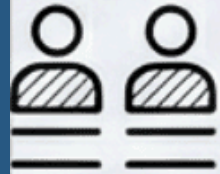
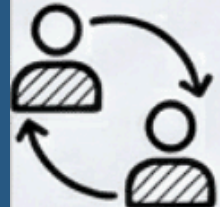
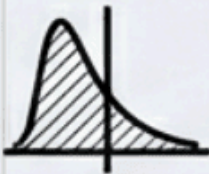
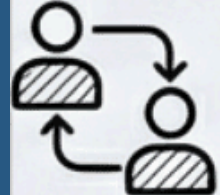
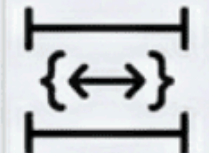
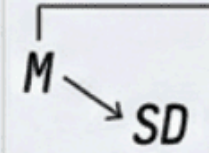


Remember

- You generated this data
- Every number came from something **you** did
- The framework helps you tell the story

See You Thursday!

Research Terminology Cheat Sheet

DESIGNS	HYPOTHESES	ANALYSIS KEYS
 Between-subjects: Different participants.	 Null: No effect (=).	 Parametric: Standard distribution.
 Within-subjects: Same participants.	 Experimental: Effect exists ($\neq$).	 Non-parametric: Skewed data.
 Matched: Paired on characteristics.	 One-tailed: Direction known.	 Mean: Average.
	 Two-tailed: Direction unknown.	 SD: Spread.
		 APA: $M = x.xx, SD = x.xx$